

40. A bus covers the first half of a certain distance with speed v_1 and the second half with a speed v_2 . The average speed during the whole journey is :

- (a) $\frac{v_1 v_2}{v_1 + v_2}$ (b) $\frac{2v_1 v_2}{v_1 + v_2}$
(c) $\frac{v_1 + v_2}{2}$ (d) $\sqrt{v_1 v_2}$

U.P.P.C.S. (Pre) 2024

Ans. (b)

Let the distance covered by the bus = d

$$\text{First half of the distance} = \frac{d}{2}$$

$$\text{Second half of the distance} = \frac{d}{2}$$

$$\text{Time taken to cover the first half } t_1 = \frac{\text{distance}}{\text{speed}} = \frac{d/2}{v_1} = \frac{d}{2v_1}$$

$$\text{Time taken to cover the second half } t_2 = \frac{d/2}{v_2} = \frac{d}{2v_2}$$

$$\begin{aligned} \text{Total time taken during the whole journey } t &= t_1 + t_2 \\ &= \frac{d}{2v_1} + \frac{d}{2v_2} \\ &= \frac{d(v_1 + v_2)}{2v_1 v_2} \end{aligned}$$

$$\begin{aligned} \therefore \text{Average speed during the whole journey} &= \frac{\text{Total distance}}{\text{Total time}} = \frac{d}{t} \\ &= \frac{d}{\frac{d(v_1 + v_2)}{2v_1 v_2}} \\ &= \frac{2v_1 v_2}{v_1 + v_2} \end{aligned}$$

41. What is the correct equation for finding the acceleration?

- (a) $a = \frac{v - u}{t}$ (b) $a = u + vt$
(c) $a = \frac{v + u}{t}$ (d) $a = \frac{v + u}{2}$

U.P.P.S.C. (Pre) 2003

Ans. (a)

As we know $v = u + at$ or $at = v - u$

$$\therefore a = \frac{v - u}{t}$$

where, v = final velocity of the particle

u = initial velocity of the particle

a = acceleration, t = time

42. The working principle of a washing machine is :

- (a) Centrifugation (b) Dialysis
(c) Reverse osmosis (d) Diffusion

I.A.S. (Pre) 1997

U.P.P.C.S. (Mains) 2004

Uttarakhand Lower Sub. (Pre) 2010

67th B.P.S.C. (Pre) (Re. Exam) 2022

Ans. (a)

The working principle of a washing machine is based on centrifugation. Centrifugation is a technique which involves the application of centrifugal force to separate particles from a solution according to their size, shape, density, viscosity of the medium and rotor speed.

43. The separation of cream from milk by churning is due to –

- (a) Gravitational force (b) Cohesive force
(c) Centrifugal force (d) None of the above

Jharkhand P.C.S. (Pre) 2013

Ans. (c)

The separation of cream from milk by churning is due to centrifugal force.

Motion Under Gravity

Notes

The Motion of Planets :

- All the planets including the earth, move around the sun in their orbits.
- A natural satellite is a celestial body in space that orbits around planets.
- Mercury and Venus planets have no natural satellites.
- Earth has one natural satellite.
- Mars has two little satellites.
- Jupiter has 97 known satellites with confirmed orbits (As of 30 April, 2025).
- Saturn has maximum numbers of known satellites with confirmed orbits i.e. 274 (As of 30 April, 2025).
- As of 30 April, 2025, Uranus has 28 satellites, Neptune has 16 satellites, and dwarf planet Pluto has 5 satellites.

Kepler's Laws of Planetary Motion :

- Johannes Kepler, a German astronomer, propounded three laws about planetary motion around the sun.
- These laws are as follows :

First Law –

- All planets move around the sun in an elliptical orbit, having the sun at one of the two foci.

Second Law –

- A radius vector joining any planet to the sun sweeps out equal areas in an equal length of time.
- The second law clears that closest planet to the sun moves fastest and farthest planet moves slowest.

Third Law –

- The square of the orbital period of a planet is directly proportional to the cube of the semi-major axis of its orbit.
- Kepler's third law states $P^2 \propto a^3$. It is a mathematical formula. It means that if you know the planet orbit (P = how long it takes the planet to move around the sun) then you can determine that planet distance from the sun (a = the semi-major axis of the planet's orbit).

Newton's Law of Gravitation :

- Newton had an opinion that all the bodies of the Universe attract each other by a force.
- This ubiquitous force is called the gravitational force.
- On this basis, Newton propounded following law about gravitation –

"The attracting force between two objects is directly proportional to the product of their masses and inversely proportional to the square of the distance between their centres."

- Newton's Law of Gravitation can be expressed mathematically as follows :

If two objects whose masses are m_1 and m_2 respectively, and they are positioned at a distance r , then the attraction of force between them will be

$$F \propto \frac{m_1 m_2}{r^2}$$

$$\Rightarrow F = G \frac{m_1 m_2}{r^2}$$

where G is a proportional constant known as Gravitational constant.

- If the objects mass is unit and distance between them is unit, then –

$$F = G \frac{1 \times 1}{1 \times 1}$$

$$F = G$$

It means that Gravitational constant is the force of attraction between two objects with unit mass and positioned at a unit distance.

- In SI System its value is 6.674×10^{-11} newton.metre²kg⁻².
- Its value was first calculated by scientist Cavendish.

Gravity and Gravitational acceleration of Earth :

- Gravity is attraction force by which Earth attracts any object towards its centre.
- It is a peculiar form of gravitation.
- When an object is thrown or dropped freely, it falls towards the Earth due to the gravity of Earth and its falling

rate is continuously increasing due to the production of acceleration, which is known as gravitational acceleration. It is represented by 'g'. Its unit is metre/second².

Relation between 'g' & 'G' :

- Gravitational acceleration of Earth $g = \frac{G M_e}{R_e^2}$

where, G = Gravitational constant

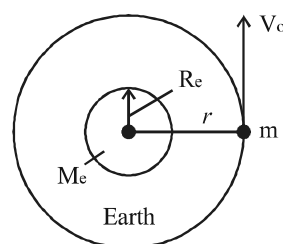
M_e = Mass of Earth

R_e = Radius of Earth

- Value of 'g' does not depend upon the mass of the object.
- If two objects of different mass are dropped freely (in absence of air) from the same height, equal acceleration will produce in both objects, they will reach at the same time to surface of the earth.
- Although, in the presence of air their acceleration will vary due to upthrust force and viscosity.
- In this condition, the heavy object will reach prior on earth.
- The value of 'g' differs at different places of Earth surface.
- The value of 'g' is minimum at equator and maximum at poles.
- The value of 'g' also varies by going upward or downward from the earth surface. In both cases, it decreases.

Circular Motion of Satellites :

- When a satellite revolves around Earth in an orbit, a centripetal force acts on it.
- This force is gravitational force imposed by Earth on satellite.



If a satellite of m mass is orbiting around the Earth with r radius and by V_o speed, centripetal force on satellite will be –

$$= \frac{m V_o^2}{r}$$

$\therefore$ gravitational force is centripetal force

$$\therefore \frac{G M_e m}{r^2} = \frac{m V_o^2}{r} \quad \dots\dots\dots (i)$$

$$\Rightarrow V_o = \sqrt{\frac{G M_e}{r}}$$

where G = Gravitational constant

M_e = Mass of Earth

we know that,

$$g = \frac{G M_e}{R_e^2}$$

$$\Rightarrow G M_e = g R_e^2 \quad \dots\dots\dots (ii)$$

By equation (i) & (ii)

$$V_o = \sqrt{\frac{g R_e^2}{r}}$$

$$V_o = R_e \sqrt{\frac{g}{r}}$$

Escape Velocity :

- Generally when we throw an object above, it after reaching to a certain height, comes down due to the gravitational force of Earth.
- Escape velocity is the minimum speed needed for a free object to escape from the gravitational influence of Earth and object never comes back to Earth.
- The kinetic energy required for escape velocity is known as escape energy.

if Mass of Earth = M_e

Radius of Earth = R_e

$$\text{Then, Escape velocity } V_e = \sqrt{\frac{2 G M_e}{R_e}}$$

where G = Gravitational constant

From the above equation, it is clear that escape velocity does not depend upon the mass of the object.

- If the gravitational acceleration of Earth is 'g'

$$\text{then, } g = \frac{G M_e}{R_e^2}$$

$$\square G M_e = g R_e^2$$

$$\backslash V_e = \sqrt{\frac{2 g R_e^2}{R_e}}$$

Gravitational acceleration (g) = 9.8 m/second^2

Radius of Earth = $6.37 \times 10^6 \text{ metre}$

By placing these values in equation –

$$V_e = \sqrt{2 \times 9.8 \times 6.37 \times 10^6} = 11.2 \times 10^3 \text{ metre/second}$$

$$= 11.2 \text{ km/second}$$

- It is clear that if any object is thrown above the surface of the earth at the minimum speed of 11.2 km/second , that object will move in space and never comes back to the Earth.

Circular Motion and Escape Velocity relation :

- If the circular motion of a satellite revolving nearby to Earth is V_o and the escape velocity of an object thrown from earth surface is V_e , then–

$$V_e = \sqrt{2} V_o$$

- It is clear that a satellite revolving nearby to earth, if by any reason, its speed exceeds by $\sqrt{2}$ (or about 1.414) times, it will escape into space, leaving its orbit.

Simple Periodic Motion :

- When an object moves both sides in a straight line from its equilibrium state, then this motion is called simple periodic motion, e.g. the movement of an object hanged with spring and motion of the simple pendulum.

Simple Pendulum –

- An ideal pendulum consists of a point mass suspended by a weightless, inextensible, perfectly flexible thread and free to vibrate without friction.
- Applied pendulum in the lab is a solid sphere of any metal tight with light & thin thread with a rigid base.
- The solid sphere of metal is called bob, and the length between point of suspension and gravity centre of the sphere is called effective length.

Periodic Time of Simple Pendulum –

- Periodic Time of any simple pendulum is represented by following formula :

$$T = 2\pi \sqrt{l/g}$$

where l = effective length of simple pendulum

g = gravitational acceleration

- From the above formula, it is clear that periodic time does not depend upon the mass of the sphere.
- Time period (T) of the pendulum is directly proportional to under root of effective length.
- If the effective length of pendulum increases by four times, periodic time will be increased by two times.
- Time period (T) of the pendulum is inversely proportional to under root of gravitational acceleration (g).

For example - when a pendulum clock is carried on the mountain or in any mine, 'g' value decreases, time period increases and the clock becomes slow.

- The time period of a simple pendulum whose length tends to infinity is 84.6 minutes. It is the maximum limit of a simple pendulum.
- A second pendulum is a pendulum whose time period is precisely 2 seconds.

Question Bank

1. Who among the following anticipated Newton by declaring that all things gravitate to the Earth?

(a) Aryabhatta (b) Varahamihira
(c) Buddhagupta (d) Brahmagupta

I.A.S. (Pre) 1995

Ans. (d)

Brahmagupta was a great 7th century AD mathematician and astronomer. He was born in Bhinmal in Rajasthan. He is best known for his 'Brahmasphuta Siddhanta' in which he declared that all things gravitate to the Earth.

2. Who gave the theory of gravity?

(a) Charles Newton (b) Charles Babbage
(c) Isaac Newton (d) John Adams

M.P.P.C.S. (Pre) 2010

Ans. (c)

In 1687, Isaac Newton published his law of gravitation in his book "Mathematical Principles of Natural Philosophy". Newton proposed that everybody in the universe is attracted to every other body with a force that is directly proportional to the product of their masses and inversely proportional to the square of the distance between them. In term of mathematical relationships, Newton's universal law of gravitation states that the Gravitational Force (F) between two particles of mass m_1 and m_2 has a magnitude of :

$$F = G \frac{m_1 m_2}{r^2}$$

where, r = the distance between both the masses,
 G = Gravitational constant.

3. Who propounded the universal law of gravitation?

(a) Newton (b) Galileo
(c) Copernicus (d) Einstein

M.P.P.C.S. (Pre) 2000

Uttarakhand U.D.A./L.D.A. (Pre) 2007

Ans. (a)

See the explanation of above question.

4. Astronauts in space can not stand erect. This is because, in space –

(a) There is no gravity
(b) Viscous forces of the atmosphere are very strong
(c) Solar wind exert an upward force
(d) Atmospheric pressure is very low

U.P.P.C.S. (Mains) 2007

Ans. (a)

In space, astronauts can't stand erect because there is no gravity or microgravity. In this condition people or objects appear to be weightless. Microgravity affects the human body in several ways. For example muscles and bones can become weaker without gravity.

5. If an apple is released from an orbiting spaceship, it will –

(a) Fall towards the Earth
(b) Move at a lower speed
(c) Move along with the spaceship at the same speed
(d) Move at a higher speed

U.P.P.C.S. (Pre) 2010

Ans. (c)

If an apple is released from an orbiting spaceship, it will move along with the spaceship at the same speed.

6. The Leaning Tower of Pisa does not fall because –

(a) It is tapered at the top
(b) It covers a large base area
(c) Its centre of gravity remains at the lowest position
(d) The vertical line through the centre of gravity of the tower falls within its base

U.P.P.C.S. (Pre) 2009

Ans. (d)

Anything will remain stable as long as the vertical line through the centre of gravity falls within its base. The centre of gravity is that point where the whole mass of the body is supposed to be concentrated. Till today the vertical line from the centre of gravity has been falling within the base of this tower. That is why it has not fallen. It is believed that when the tower leans further and the line from its centre of gravity passes out of its base, it will fall down.

7. If the distance between the Earth and the Sun were twice what it is now, the gravitational force exerted on the Earth by the Sun would be –

(a) Twice as large as it is now
(b) Four times as large as it is now
(c) One fourth of what it is now
(d) Half of what it is now

U.P.P.C.S. (Mains) 2009

Ans. (c)

Let the present distance between the Earth and the Sun is 'r' then according to the Newton's universal law of gravitation, the gravitational force imposed by the Sun on the Earth :

$$F_1 = G \frac{M_s M_e}{r^2} \dots\dots\dots(1)$$

M_s = Mass of Sun

M_e = Mass of Earth

G = Gravitational constant

If the distance between the Earth and the Sun were twice, then,

$$F_2 = G \frac{M_s M_e}{(2r)^2} = G \frac{M_s M_e}{4r^2} \dots\dots\dots(2)$$

from equation (1) and (2)

$$\frac{F_1}{F_2} = \frac{4}{1} \Rightarrow 4F_2 = F_1$$

$$\therefore F_2 = \frac{1}{4} F_1$$

8. If an object is falling from above, its weight remains –

- (a) Zero (b) Same
(c) Change (d) None of the above

M.P. P.C.S. (Pre) 1992

Ans. (b)

In ideal conditions (when air resistance is negligible) if an object is falling from above, its weight remains same because mass of the object and the gravity remains unchanged.

9. If a feather, a rubber ball and a wooden ball are falling freely simultaneously from the same height in vacuum, then :

- (a) the feather will reach at the ground first
(b) the rubber ball will reach at the ground first
(c) the wooden ball will reach at the ground first
(d) all the three will reach at the ground together
(e) None of the above/More than one of the above

67th B.P.S.C. (Pre) 2022

Ans. (d)

The feather, rubber ball and wooden ball will fall with the same speed and reach the ground together, because there is no air resistance in the vacuum, and gravitational force acts same on all the objects. Hence, every object undergoes free fall in the vacuum with the same acceleration.

10. If pieces of equal size of woods, wax and iron falling from a same height, which one will reach the surface of the earth first?

- (a) Wood (b) Wax
(c) Iron (d) Reach at the same time

R.A.S./R.T.S. (Pre) 1992

Ans. (d)

In ideal conditions (when air resistance is negligible) all pieces of equal size of woods, wax and iron falling from same height despite their weight difference will hit the ground at the same time because the same gravitational acceleration (g) applies to all objects.

11. The two balls of iron and wood of same weight dropped from the same height in air, then –

- (a) They reach the ground (earth) at the same time.
(b) One falls first, another falls later.
(c) The wooden ball falls later.
(d) They reach the ground after some gap.

U.P.P.C.S. (Pre) 1992

Ans. (c)

If there is no resistance from the air, both the balls will hit the ground at the same time. But in the presence of air, the iron ball falls first because of its smaller size and wood ball falls later because of its larger size. The resistance from air slows down the fall of larger thing with same weight. So, option (c) will be the correct answer.

12. Two balls A and B are of 10 kg. and 1 kg. respectively. They are dropped simultaneously from a height of 20 m. Which of the following is correct?

- (a) A will reach the ground first
(b) B will reach the ground first
(c) Both A and B will reach the ground at the same time
(d) None of the above

U.P. Lower Sub. (Pre) 2015

Ans. (c)

Both balls A and B will reach the ground at the same time. Because both balls are thrown from the equal height of 20 metre.

Thus, $h = 20\text{m}$

Initial velocity $u = 0$

speed equation (for gravitational velocity)

$$V^2 = u^2 + 2gh \quad [V \rightarrow \text{final velocity}]$$

here, $V^2 = 2gh$ ($\because u = 0$)

$$\text{or } V = \sqrt{2gh}$$

Since the height of both balls are equal ($h = 20$) and the formula of final velocity ($V = \sqrt{2gh}$) is not dependent on the mass of the balls. Thus, both the balls will reach the ground at the same time.

13. Two objects of different masses falling freely near the surface of the Moon would :

- (a) have different accelerations
(b) undergo a change in their inertia
(c) have same velocity at any instant
(d) experience forces of same magnitude

69th B.P.S.C. (Pre) 2023

Ans. (c)

The question does not give the parameters for height and time. Assuming that both objects are dropped at the same time from the same reasonable height, they will have 'the same velocity at any instant' as they fall freely near the surface of the moon.

14. Assertion (A) : The weight of human being on the Moon is 1/6 in comparison to Earth.

Reason (R) : The Moon does not have gravity like Earth.

Select the answer from following codes –

- (a) Both (A) and (R) are true, and (R) is the correct explanation of (A).

- (b) Both (A) and (R) are true, but (R) is not the correct explanation of (A).
 (c) (A) is true, but (R) is false.
 (d) (A) is false, but (R) is true.

Chhattisgarh P.C.S. (Pre) 2008

Ans. (c)

The Moon is the Earth's only natural satellite. The surface gravity of the Moon is only $1/6$ that of the Earth. So the weight of human being on the Moon will be $1/6$ of that on the Earth. Hence, assertion is true, but reason is false.

15. The weight of anybody is –

- (a) Same at every place on Earth's surface
 (b) Maximum at poles
 (c) More on equator
 (d) More on mountains than in planes

U.P.P.C.S. (Mains) 2009

U.P.P.C.S. (Pre) 2006

Ans. (b)

At a given place, the value of acceleration due to gravity is constant but it varies from one place to another place on the Earth's surface. It is due to this fact that Earth is not a perfect sphere. It is flattened at the poles and bulges out at the equator (ellipsoidal shaped).

The polar radius R_p is less than the equatorial radius R_e . Now,

$$g = \frac{GM}{R^2}$$

Now, as 'G' & 'M' remain constant, therefore

$$g \propto \frac{1}{R^2}$$

Thus the value of 'g' is minimum at equator and maximum at the poles. It means 'g' increases as we move from the equator to poles.

16. The weight of human body is –

- (a) Maximum at the poles.
 (b) Same at every place on the Earth's surface.
 (c) Maximum at the equator.
 (d) More on the mountains than planes.

U.P.P.C.S. (Mains) 2015

Ans. (a)

See the explanation of above question.

17. If the spinning speed of the Earth increases, then the weight of the body at the equator will :

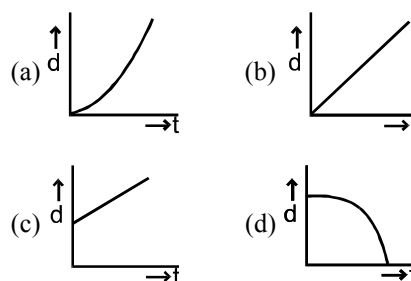
- (a) increase (b) decrease
 (c) remain same (d) be doubled
 (e) None of the above/More than one of the above

67th B.P.S.C. (Pre) (Re. Exam) 2022

Ans. (b)

The weight of a body on the Earth is the amount of gravitational pull on it exerted by the Earth. Due to Earth's rotation, centrifugal force also exerted on a body on Earth, which tries to move the body away from the centre of the Earth. At the equator, centrifugal force is maximum as the velocity of rotation is maximum at the equator of the Earth. Thus, at the equator, centrifugal force acts against gravity to reduce the weight of the body. With the increase in the speed of rotation of the Earth, the centrifugal force at equator will increase and the weight of the body will seem to have decreased.

18. The variation of displacement (d) with time (t) in the case of a particle falling freely under gravity from rest is correctly shown in graph :

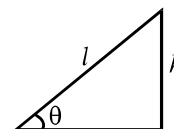


I.A.S. (Pre) 1996

Ans. (a)

For an object falling freely from the state of rest under the gravitational acceleration, the equation of motion is $d = \frac{1}{2}gt^2$ ($\because d = ut + \frac{1}{2}gt^2$ and $u = 0$), which is also an equation of the parabola and $d \propto t^2$. Therefore option (a) is correct.

19. A smooth plane inclined at an angle θ with the horizontal as shown in the given figure. A body starts from rest and slides down on the inclined surface. The time taken by the body to reach the bottom is –



- (a) $\sqrt{\frac{2h}{g}}$ (b) $\sqrt{\frac{2l}{g}}$
 (c) $\frac{1}{\sin \theta} \sqrt{\frac{2h}{g}}$ (d) $\sin \theta \sqrt{\frac{2h}{g}}$

I.A.S. (Pre) 1997

Ans. (c)

The body is placed on the smooth plane inclined at an angle θ , thus it moves under $g \sin \theta$ instead of g and we know that it starts moving from rest, therefore initial speed $u = 0$

The equation of motion $s = ut + \frac{1}{2}gt^2$

Hence $l = \frac{1}{2}g \sin \theta \cdot t^2$

or, $\frac{2l}{g \sin \theta} = t^2 \dots (i)$

Now as we know $\sin \theta = \frac{h}{l} \Rightarrow l = \frac{h}{\sin \theta}$

putting this value of l in equation (i), we get

$$t^2 = \frac{2 \times h}{g \sin^2 \theta}$$

$$\Rightarrow t = \frac{1}{\sin \theta} \sqrt{\frac{2h}{g}}$$

- 20. A ball is dropped from the top of a high building with a constant acceleration of 9.8 m/s^2 . What will be its velocity after 3 seconds ?**

- (a) 9.8 m/s (b) 19.6 m/s
(c) 29.4 m/s (d) 39.2 m/s

I.A.S. (Pre) 1998

Ans. (c)

According to question,

$a = 9.8 \text{ m/s}^2$, $t = 3 \text{ sec.}$, $u = 0$

we know that, $v = u + at = 0 + 9.8 \times 3 = 29.4 \text{ m/s.}$

- 21. If the radius of the Earth were to shrink by one percent, its mass remaining the same, the value of g on the Earth's surface would :**

- (a) Increase by 0.5% (b) Increase by 2%
(c) Decrease by 0.5% (d) Decrease by 2%

I.A.S. (Pre) 2003

Ans. (b)

We know that $g = \frac{GM}{R^2} \dots (i)$

After shrinking by 1% new radius will be $0.99 R$

Then, $g' = \frac{GM}{(0.99R)^2} \dots (ii)$

From equation (i) and (ii)

$$\frac{g'}{g} = \frac{R^2}{(0.99R)^2} = \frac{1}{(0.99)^2} = 1.02$$

□ $g' = 1.02g$

Thus, the value of g would increased by 2%.

- 22. The acceleration due to gravity of a destructive earthquake will be –**

- (a) $> 550 \text{ cm/sec}^2$ (b) $> 750 \text{ cm/sec}^2$
(c) $> 950 \text{ cm/sec}^2$ (d) $> 980 \text{ cm/sec}^2$

I.A.S. (Pre) 1994

Ans. (d)

The acceleration due to gravity of a catastrophic earthquake will be greater than 980 cm/sec^2 or 9.8 m/sec^2 .

- 23. The mass of a body on the Earth is 100 kg (acceleration due to gravity, $g_e = 10 \text{ m/s}^2$). If acceleration due to gravity on the Moon is ($g_m/6$), then the mass of the body on the Moon is :**

- (a) $100/6 \text{ kg}$ (b) 60 kg
(c) 100 kg (d) 600 kg

I.A.S. (Pre) 2001

Ans. (c)

The mass of a body does not change with respect to gravity. It is the weight that changes due to increase or decrease in gravity. So the mass of body will remain the same as on the Moon as it was on the Earth.

- 24. Weightlessness is –**

- (a) Zero gravity (b) When gravity decrease
(c) In vacuum (d) None of the above

M.P.P.C.S. (Pre) 1991

Ans. (a)

The phenomenon of weightlessness occurs when there is no force to support on your body. When your body is effectively in 'free fall', accelerating downwards at the acceleration of gravity, then you are not being supported. The sensation of apparent weight comes from the support that you feel from the floor, from the seat etc. The term 'zero gravity' is often used to describe such condition.

- 25. The Earth travels in its orbit at a speed of approx 4,400 km per hour. Why do we not feel this high speed ?**

- (a) Because we are too small compared to the size of the earth.
(b) Our relative speed with respect to the Earth along the earth's orbit is zero.
(c) The solar system as a whole is also moving.
(d) The gravity of the Earth constantly pulls us towards the Earth's centre.

U.P.P.C.S. (Mains) 2013

Ans. (b)

The Earth spins and moves through space in an orbit around the sun. We don't 'feel' this movement because we are spinning and moving through space with the Earth. Since we are standing on the Earth, we move as much (and as fast) as the Earth moves. One way to recognize the Earth's movement is to look at objects that are not attached to Earth, like the Sun

or stars. For example, if we are in a car which is moving at a constant speed on a smooth surface, we will not feel its motion. However, when the car accelerates or when the brakes are applied, we do feel the motion.

26. If the gravitational force of the Earth suddenly disappears, which are of the following will be the correct consequence?

- (a) The weight of an object will become zero but the mass will remain the same.
- (b) The mass of the object will become zero but the weight will remain the same.
- (c) Both the mass and the weight of the object will become zero.
- (d) The mass of the object will increase.

U.P.P.C.S. (Mains) 2012

Ans. (a)

If the gravitational force of the Earth suddenly disappears, then the weight of an object will become zero but the mass will remain the same.

27. An artificial satellite orbiting around the Earth does not fall down. This is so because the attraction of Earth –

- (a) Does not exist at such distance
- (b) Is neutralized by the attraction of the Moon
- (c) Provides the necessary speed for its steady motion
- (d) Provides the necessary acceleration for its motion

I.A.S. (Pre) 2011

Ans. (d)

An artificial satellite orbiting around the Earth does not fall down because the gravitational attraction of the Earth provides the necessary centripetal acceleration for its motion in a curved path.

28. Satellite is kept moving in its orbit around the Earth; it is due to –

- (a) Centrifugal force
- (b) Centripetal force
- (c) Gravitational force or lack of it
- (d) Some other forces

U.P.P.C.S. (Mains) 2013

Ans. (b)

Centripetal force is a real force that counteracts the centrifugal force and prevents the object from 'flying out,' keeping it moving instead with a uniform speed along a circular or elliptical path. Gravity is the centripetal force that keeps planets moving around the sun and satellites moving around planets.

29. A Geosynchronous satellite continuously active in its orbit due to centripetal force which is obtained by-

- (a) The rocket engine that propelled the satellite.
- (b) The gravitational force on the satellite by the earth.

- (c) The gravitational force on the satellite by the sun.
- (d) The gravitational force on the earth by satellite.

U.P.P.C.S. (Pre) 2006

Ans. (b)

See the explanation of above question.

30. The known forces of nature can be divided into four classes, viz. gravity, electromagnetism, weak nuclear force and strong nuclear force. With reference to them which one of the following statements is not correct?

- (a) Gravity is the strongest of the four
- (b) Electromagnetism acts only on particles with an electric charge
- (c) Weak nuclear force causes radioactivity
- (d) Strong nuclear force holds protons and neutrons inside the nucleus of an atom

I.A.S. (Pre) 2013

Ans. (a)

Gravitational force is the weakest force among the four fundamental forces while strong nuclear force holds the position of strongest force among them. Hence, statement of option (a) is incorrect, while statements of other three options are correct.

31. If a person sitting in a lift then when will he feel that his weight gets increased?

- (a) When the elevator going upward expeditiously
- (b) When the elevator going down expeditiously.
- (c) Going upward with constant velocity.
- (d) Going downward with constant velocity.

U.P.P.C.S. (Pre) 1990

Ans. (a)

Let the mass of the person is 'm' and weight is 'w' and the elevator going upward with acceleration 'a'. The force acting on person–

1. the force due to gravity = mg

2. the force giving the acceleration = ma

According to Newton's second law–

$$w - mg = ma$$

$$w = m(g + a)$$

So, it is clear that when the elevator is going upward expeditiously a person actually feels a little heavier than his usual weight and on the other hand when elevator accelerates downward then a person feels a little reduction than his usual weight.

32. The time period of a pendulum –

- (a) Depends on the mass
- (b) Depends on its length

- (c) Depends on time
(d) Depends on temperature

47th B.P.S.C. (Pre) 2005

Ans. (b)

The time period of a pendulum depends on its length. As we know that $T = 2\pi\sqrt{\frac{l}{g}}$, thus the result is that the one variable that affects the period of the pendulum is the length of the string. Increases in the length lead to increases in the period.

33. Why do pendulum clocks usually lose time in summer?

- (a) Due to longer days in summer.
(b) Due to friction caused by coils.
(c) The length of the pendulum increases, increasing the time period of unit oscillation.
(d) The weight of the pendulum change in summer.

U.P.P.C.S. (Pre) 2012

U.P.P.C.S. (Pre) 1994

Ans. (c)

Time period of simple pendulum $T = 2\pi\sqrt{\frac{l}{g}}$

where l = length of the pendulum

In summer, the length of the pendulum is slightly increased. Due to this the time-period also increased. This results that the pendulum clocks are slow in summer.

34. A girl is swinging on a swing in sitting position. When the girl stands up, the period of the swing will :

- (a) Be shorter
(b) Be longer
(c) Depends on the height of the girl
(d) Not change

I.A.S. (Pre) 1997

Ans. (a)

A girl is swinging on a swing in sitting position but when she stands up, she is effectively moving her centre of gravity higher, thus shortening the effective length of the pendulum. Shortening of this length will decrease the period of the swing that means the period of the swing will be shorter.

35. Consider the following statements :

A Simple Pendulum is set into oscillation

Then :

1. the acceleration is zero when the bob passes through the mean position.
2. in each cycle the bob attains a given velocity twice.
3. both acceleration and velocity of the bob are zero when it reaches its extreme position during the oscillation.

4. the amplitude of oscillation of the simple pendulum decreases with time.

Which of these statements are correct ?

- (a) 1 and 2
(b) 3 and 4
(c) 1, 2 and 4
(d) 2, 3 and 4

I.A.S. (Pre) 2001

Ans. (c)

When an object, which is in simple periodic motion passes through its mean position, then

- (i) No force acts on it.
(ii) Its acceleration becomes zero.
(iii) It has maximum velocity.
(iv) It has maximum kinetic energy.
(v) Its potential energy is zero.

When the object reaches its extreme position then,

- (i) Its acceleration is maximum.
(ii) Counterforce acting on it is maximum.
(iii) Its Kinetic energy is zero.
(iv) It has maximum potential energy.
(v) It has zero velocity.

In each rotation, the pendulum twice attains a specific velocity. In addition, under normal circumstances oscillation of usual pendulum decreases with time (due to air resistance etc.). Thus the statement 1, 2 and 4 are correct while statement 3 is wrong.

36. A pendulum clock can run faster in –

- (a) Summer
(b) Winter
(c) Spring season
(d) Rainy season

R.A.S./R.T.S. (Pre) 1997

Ans. (b)

A pendulum clock runs faster in winter because in winter the length of the pendulum or the swing become smaller due to contraction and the clock begin to run fast.

37. Earth's escape velocity is –

- (a) 15.0 km/sec.
(b) 21.1 km/sec.
(c) 7.0 km/sec.
(d) 11.2 km/sec.

R.A.S./R.T.S. (Pre) 1993

Ans. (d)

The term escape velocity means the minimum speed that a moving object must have to escape from the gravitational field of a celestial body and move outward into space. The escape velocity of Earth is 11.2 km/sec.

38. What will happen if an object is thrown into space with the speed of 8 km/sec.?

- (a) The object will go into space
(b) It will return to earth